

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. You must simplify your answer if possible.

Problem 1. (8 points) Find all equilibrium values of the given system of the differential equations

$$\frac{dx}{dt} = -x - xy^2$$

$$\frac{dy}{dt} = -y - yx^2$$

$$\frac{dz}{dt} = 1 - z + x^2$$

Solution:
$$\begin{cases} -x - xy^2 = 0 \Rightarrow -x(1+y^2) = 0 & \text{--- ①} \\ -y - yx^2 = 0 \Rightarrow -y(1+x^2) = 0 & \text{--- ②} \\ 1 - z + x^2 = 0 \Rightarrow z = 1 + x^2 & \text{--- ③} \end{cases}$$

$$\begin{cases} \text{①} \& 1+y^2 \neq 0 \Rightarrow x=0 \\ \text{②} \& 1+x^2 \neq 0 \Rightarrow y=0 \\ \& \text{③} \end{cases} \Rightarrow z=1.$$

Thus, the equilibrium value of this differential equation is only $(0, 0, 1)$

Problem 2. (12 points) Determine the stability (stable, asymptotically stable or unstable) of all solutions of the following system of differential equations

a)

$$\frac{dx}{dt} = \begin{pmatrix} -5 & -2 \\ 2 & 1 \end{pmatrix} x$$

Solution: $A = \begin{pmatrix} -5 & -2 \\ 2 & 1 \end{pmatrix}$

$$P(\lambda) = \det(A - \lambda I) = \det \begin{pmatrix} -5-\lambda & -2 \\ 2 & 1-\lambda \end{pmatrix}$$

$$= (-5-\lambda)(1-\lambda) + 4$$

$$= \lambda^2 + 4\lambda - 1 = 0$$

$$\Rightarrow \lambda = \frac{-4 \pm \sqrt{20}}{2} = -2 \pm \sqrt{5}$$

Notice that $-2 + \sqrt{5} > 0$, so that every solution of this equation is unstable.

b)

$$\frac{dx}{dt} = \begin{pmatrix} -1 & -4 & 2 \\ 0 & 0 & 3 \\ 0 & -3 & 0 \end{pmatrix} x$$

Solution: $A = \begin{pmatrix} -1 & -4 & 2 \\ 0 & 0 & 3 \\ 0 & -3 & 0 \end{pmatrix}$

$$P(\lambda) = \det(A - \lambda I) = \det \begin{pmatrix} -1-\lambda & -4 & 2 \\ 0 & -\lambda & 3 \\ 0 & -3 & -\lambda \end{pmatrix}$$

$$= (-1-\lambda)[(-\lambda)^2 + 9]$$

$$= -(\lambda+1)(\lambda^2+9)$$

$$\Rightarrow \lambda_1 = -1, \lambda_2 = 3i, \lambda_3 = -3i$$

Since the multiplicity of λ_2 & λ_3 are both 1, every solution of this equation is stable, but not asymptotically stable.